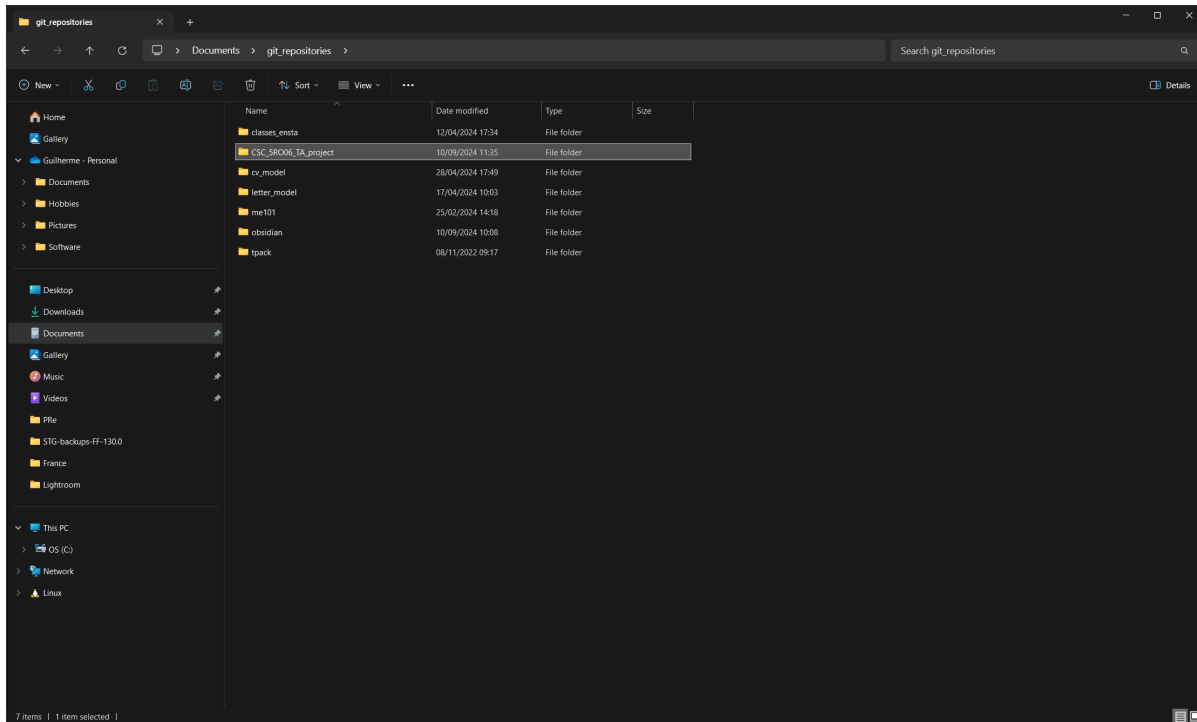


Vivado 2019.1 - Create Project

Create Folder

Create a folder where all project will be stored.

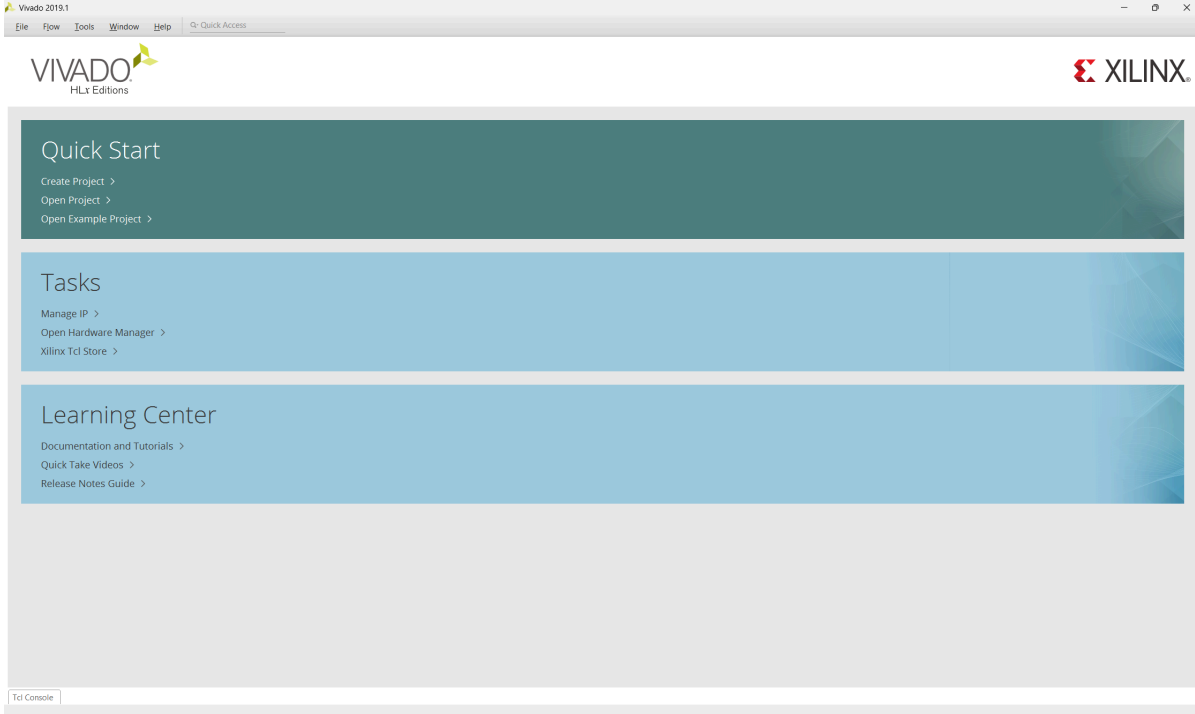


Initialize a [Git - Create Repository](#) in this recently created folder.

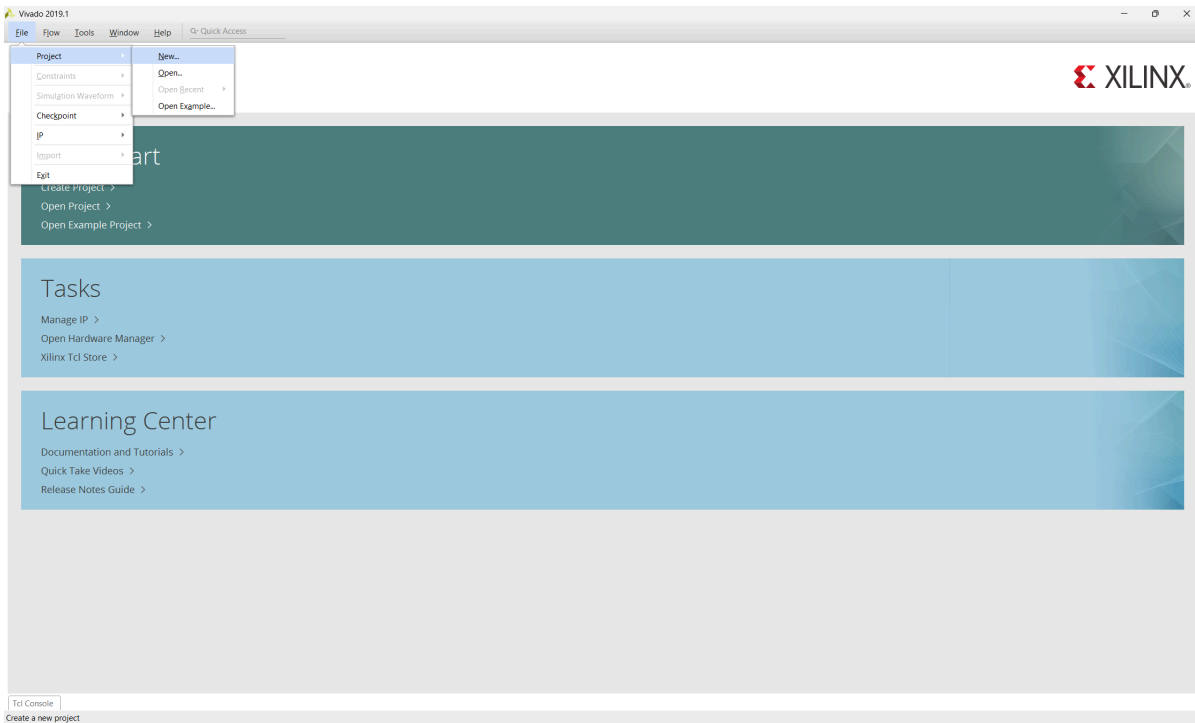
Advice

Keep all files inside recent create folder to ensure compability.

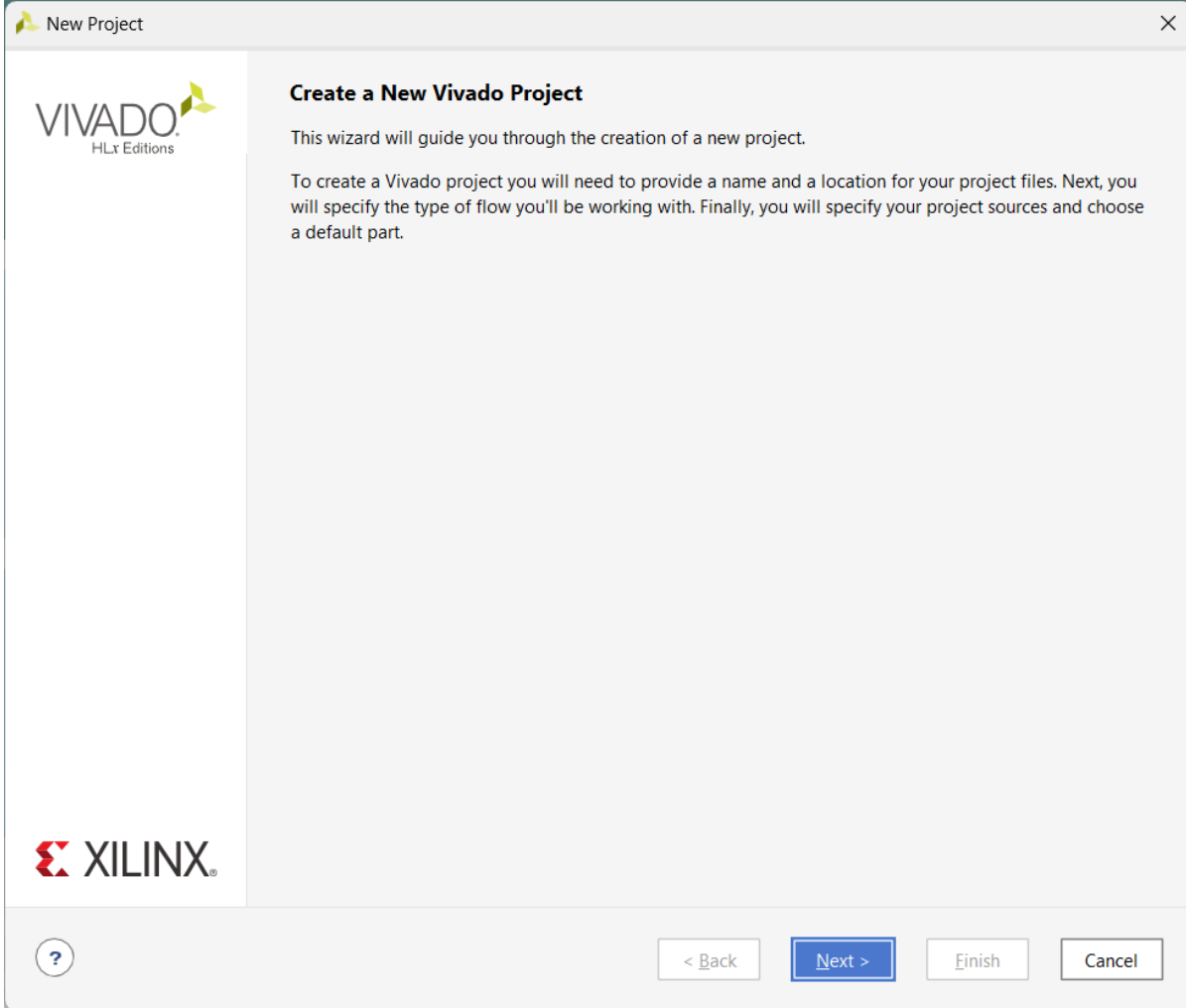
Open Vivado



Create a project under File, Project, New



Follow Project Wizard instructions



Choose folder and give project a name

New Project

Project Name

Enter a name for your project and specify a directory where the project data files will be stored.

Project name: ZedBoard_v000

Project location: C:/Users/guitr/Documents/git_repositories/CSC_5RO06_TA_project

☒ Create project subdirectory

Project will be created at: C:/.../Documents/git_repositories/CSC_5RO06_TA_project/ZedBoard_v000

? < Back Next > Finish Cancel

Select RTL Project option

New Project

Project Type

Specify the type of project to create.

☒ **RTL Project**
You will be able to add sources, create block designs in IP Integrator, generate IP, run RTL analysis, synthesis, implementation, design planning and analysis.

☐ Do not specify sources at this time

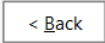
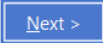
☐ **Post-synthesis Project**
You will be able to add sources, view device resources, run design analysis, planning and implementation.

☐ Do not specify sources at this time

☐ **I/O Planning Project**
Do not specify design sources. You will be able to view part/package resources.

☐ **Imported Project**
Create a Vivado project from a Synplify, XST or ISE Project File.

☐ **Example Project**
Create a new Vivado project from a predefined template.

As in many other contexts, abstraction means ‘taking away’. This is effectively the hiding of detail — the higher the level of abstraction, the more detail is hidden. There are various models for abstraction presented in the literature, many of which derive from the Gajsku and Kuhn ‘Y-chart’ for Very Large Scale Integration (VLSI) design [9], published as long ago as 1983! However, here we will focus our discussion on current FPGA design practices, and consider that there are four levels of abstraction, in ascending order: *structural*, *RTL*, *behavioural*, and *high-level*, as depicted in Figure 14.1.

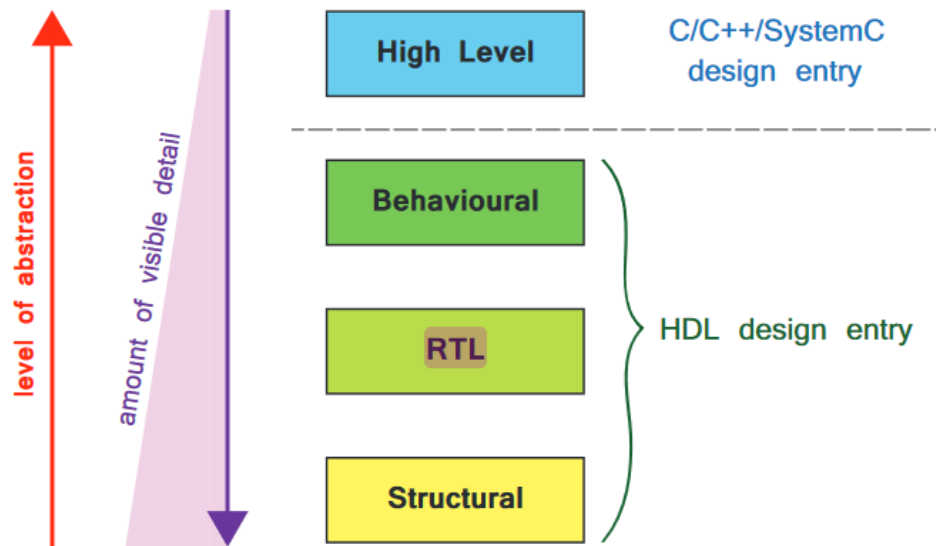


Figure 14.1: Levels of abstraction in FPGA designs

The Zynq eBook 3, page 257

More often, designers work at the ‘Register Transfer Level’ (RTL) level. This level of abstraction hides the technology-level details, but still represents a description style from which registers, and the operations occurring between registers, can be interpreted.

Logic synthesis tools are designed to work at this level of abstraction, i.e. to translate RTL code to hardware. Behavioural HDL descriptions are at a higher level of abstraction than RTL, and constitute an algorithmic description of the circuit (i.e. how it ‘behaves’), rather than an expression of individual operations with respect to registers.

New Project

Add Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those files, to add to your project. Create a new source file on disk and add it to your project. You can also add and create sources later.

+ - ↑ ↓

Use Add Files, Add Directories or Create File buttons below

Add Files

Add Directories

Create File

☐ Scan and add RTL include files into project

☐ Copy sources into project

☒ Add sources from subdirectories

Target language:

Verilog

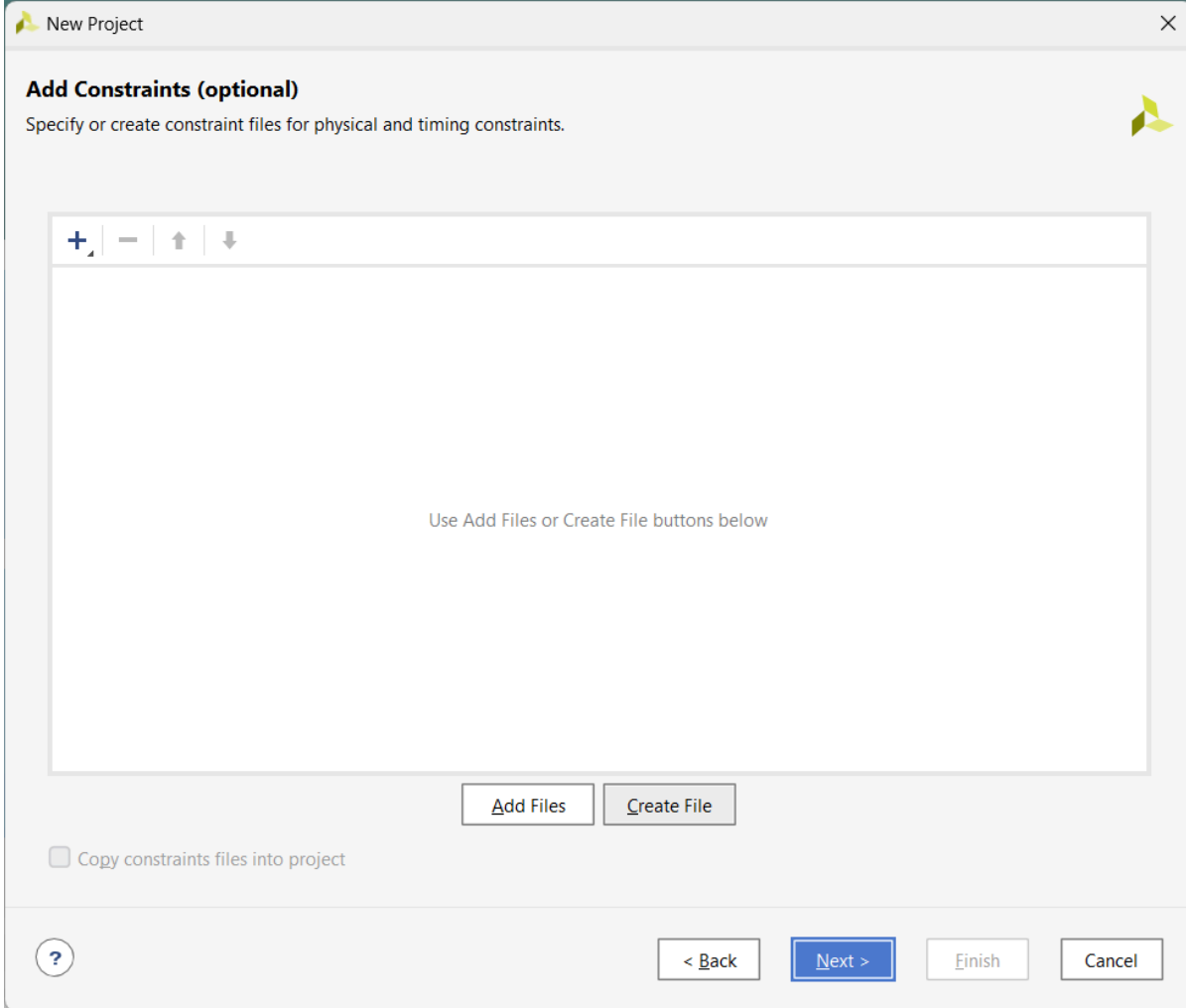
Simulator language:

Mixed

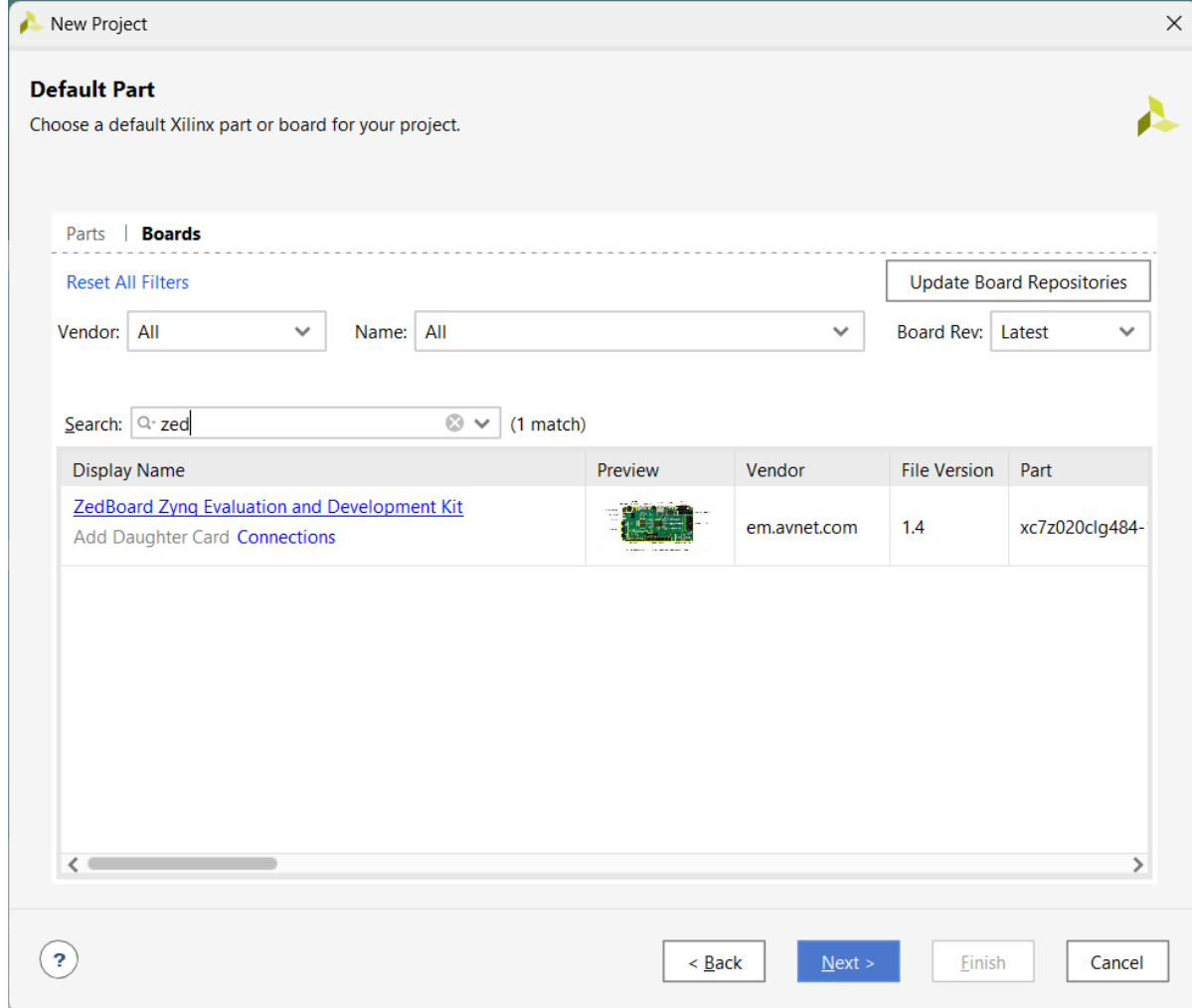
?< BackNext >

FinishCancel

Search for include files in your project directories



Search Zed on the Boards tab and select the ZedBoard option.



Project finalized.



New Project Summary

- i** A new RTL project named 'ZedBoard_v000' will be created.
- !** No source files or directories will be added. Use Add Sources to add them later.
- !** No constraints files will be added. Use Add Sources to add them later.
- i** The default part and product family for the new project:
Default Board: ZedBoard Zynq Evaluation and Development Kit
Default Part: xc7z020clg484-1
Product: Zynq-7000
Family: Zynq-7000
Package: clg484
Speed Grade: -1



To create the project, click Finish



< Back

Next >

Finish

Cancel